

Tarun Singh

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EXPERIENCE

Data Scientist

March 2025 – Present

OMR India Outsources Pvt Ltd

Bengaluru, India

- Fine-tuned and deployed a lightweight LLM for movie script summarization and genre classification, improving response accuracy by **30%** while reducing manual review effort by **35%**.
- Built a movie recommendation system for Reeloid using movie-category embeddings and cosine similarity, increasing recommendation clicks by **20%** and improving user session time by **15%**.
- Built a LangChain-based LLM solution to convert video content into quizzes by processing video → audio → transcript → questions, reducing manual question creation effort by **40%** and improving learner engagement by **15%**.
- Developed **60+** REST APIs for the Reeloid application using FastAPI and MongoDB, and optimized **100+** existing endpoints, reducing API response time by **50%** and improving overall application performance by **25%**.

AI/ML Career Transition & Upskilling

February 2024 – February 2025

Self-Directed Study & Project Development

- Completed focused upskilling in Machine Learning, Deep Learning, NLP, and Generative AI through structured study and project development.
- Built and deployed 3 end-to-end AI/ML projects spanning RAG pipelines, transformer fine-tuning, and recommendation systems.

Quality Assurance Engineer

June 2023 – February 2024

Paytm

Delhi, India

- Implemented and automated **100+** end-to-end test cases in Java within IntelliJ IDEA, integrating suites into the CI pipeline to cut manual testing effort by **30%** and accelerate release verification by **20%**.
- Designed and refactored a modular, data-driven test framework, improving test reusability and reducing flaky failures by **35%**, which contributed to a **40%** drop in production defects across critical user flows.
- Led cross-functional testing sprints with Product and Engineering to define acceptance criteria and risk-based test plans, decreasing test-case development and execution time by **29%** and shortening bug-to-fix cycles by **25%**.
- Conducted automated regression and backward-compatibility testing for weekly releases, introducing selective regression runs that reduced regression escapes by **30%** and improved post-release product stability by **25%**.

PROJECTS

YouTube Comments Sentiment Analysis | Source Code

- Developed a YouTube comment sentiment analyzer by fine-tuning MobileBERT on labeled YouTube data to classify comments into six emotions, improving accuracy by **30%** and reducing manual moderation effort by **35%**.
- Optimized inference with TensorFlow and Float16 quantization, shrinking model size by **60%** and lowering latency by **45%**.
- **Tech used:** Deep Learning, MobileBERT, Transformers, NLP, TensorFlow, TensorFlow Lite, Quantization, YouTube API.

Harry Potter In Depth Query Bot | Source Code

- Built a Harry Potter query bot using LangChain RAG with Chroma vector search, improving answer accuracy by **30%**.
- Implemented embeddings and prompt templates to reduce hallucinations, cutting incorrect responses by **25%**.
- **Tech used:** Generative AI, GPT-4o-mini, LangChain, Chroma, RAG, Embeddings, Prompt Engineering, Flask, Docker

Bollywood Song Recommendation System | Source Code

- Built a Bollywood song recommender using Spotify metadata and cosine-similarity models, boosting relevance by **20%**.
- **Tech used:** Machine Learning, Python, Scikit-learn, Flask, Docker, DVC, Git.

SKILLS

Languages & Libraries : Python, C++, TensorFlow, PyTorch, Scikit-Learn, NumPy, Pandas, LangChain, FastAPI

Tools : AWS, Docker, SQL, MongoDB, DVC, MLflow, Apache Kafka, Git / GitHub

EDUCATION

Dr. A.P.J. Abdul Kalam Technical University

August 2019 - June 2023

Bachelor of Technology in Computer Science and Engineering

Uttar Pradesh, India